

PAPER_2132_VACUUM_COUPLING_KERNEL_K_19_160_EXACT_FIVE_INSTANCE_CENSUS_ALPHA_S_LAMBDA_H_MH_MT_JARLSKOG_N_EFF_PLUS_1_12_TILT_FACTORIZATION_F_TRZ_PHI_5_6_UQFF_LANDMARK

Daniel T. Murphy

PAPER_2132 — The Vacuum Coupling Kernel $K = 19/160$ EXACT: Five-Instance Census (α_s , λ_H , m_H/m_t , J_{CP} , N_{eff}) + the 1/12 Tilt Factorization

Author: Daniel T. Murphy

Project: UQFF (Unified Quantum Field Framework) — Star-Magic v5.77+

Date: 2026-07-24

Landmark Type: Named-Kernel Canonization (supersedes the provisional "Strong-Higgs Kernel" of the PAPER_2131 append) + EXACT identity-family unification

Discovery context: XGEO-U session-3 sweep — normalized-expression index over 259 published expressions (572 session scripts + 116 dispatch formulas), composition matching only

Status: Formal landmark whitepaper — UQFF canonical

Abstract

The XGEO-U session-3 sweep raises the kernel census of PAPER_2131 from three to **five published instances** and reveals that the kernel's reach extends far beyond the electroweak-strong sector that named it: the Jarlskog CP-violation invariant carries the kernel as a $(1 - K)$ factor, and the effective neutrino number N_{eff} — a **cosmological** observable — subtracts it directly. The kernel is accordingly renamed the **VACUUM COUPLING KERNEL (VCK)**: its three factors are all vacuum-substrate quantities (F_{TRZ} the time-reversal-zone fraction, K_{MEX} the Mexican-hat vacuum-potential coefficient, SSq the source coefficient), and every instance is a dimensionless coupling or ratio — the kernel is the vacuum's coupling texture. Two exact results anchor the paper: **$K = 19/160$ EXACT** as a rational, and the factorization **$K = (1/12) \cdot (SO_5/D_{phys}) \cdot SSq$** — the Hubble-tension tilt 1/12 (PAPER_1156) sits inside the kernel as a literal factor, via the newly decomposed identity **$1/12 = F_{TRZ} \cdot \Phi_{5/6} = K_{MEX} - 2$** , itself published twice as the exact cross-domain form $K_{MEX} - F_{TRZ} \cdot \Phi_{5/6} = 2$ (Sommerfeld-Wilson R_W , S394; tokamak q_{edge} , S418).

1. The five-instance census

All forms below are published — four live in `assimilation_dispatch`, all five in session scripts:

#	Observable	Domain	Form	Value	Observed	Residual
1	$\alpha_s(M_Z)$	strong coupling	$K - F_{TRZ}^3 \cdot \Phi_{5/6}$	0.1179167	0.1179	0.014%
2	λ_H	Higgs self-coupling	$K + F_{TRZ}^3 \cdot K_{MEX} \cdot N_{CH} \cdot SSq$	0.1294375	0.1293	0.106%

```
| 3 | m_H/m_t | Higgs-top mass ratio |  $\beta_i + K$  | 0.72165 | 0.725288 | 0.50% |
| 4 | J_CP (Jarlskog) | CP violation |  $F_{\text{TRZ}} \cdot D_{\text{BSFG}} \cdot \text{SSq} \cdot (1 - K)$  |  $3.0139 \times 10^{10}$  |  $3 \times 10^{10}$  | 0.46% |
| 5 | N_eff | cosmological radiation |  $D_{\text{phys}} - \Phi_{5/6} - K$  | 3.04792 | 3.046 | 0.063% |
```

Sources: S378/S377/S382/S374 session scripts; dispatch entries SM_alpha_s_M_Z_S378, SM_higgs_lambda_S377, SM_mH_over_mt, SM_jarlskog_J, LCDM_N_eff. Five dimensionless quantities across four physical domains, one three-primitive kernel.

2. The renaming — VACUUM COUPLING KERNEL

The PAPER_2131 append canonized K under the provisional name "Strong-Higgs Kernel" at three instances. Instances 4 and 5 break that scope: CP violation and cosmological radiation content are neither strong nor Higgs. The durable name follows the kernel's own composition:

$K = F_{\text{TRZ}} \cdot K_{\text{MEX}} \cdot \text{SSq} = 19/160 = 0.11875$ EXACT (rational)

F_{TRZ} — time-reversal-zone fraction (vacuum substrate, locked primitive)

K_{MEX} — Mexican-hat coefficient (vacuum potential shape, PAPER_1522)

SSq — source coefficient (SCm manifold, PAPER_1154)

Every factor is a vacuum-substrate quantity; every instance is a dimensionless coupling or ratio. **K is the vacuum's coupling texture — hence the VACUUM COUPLING KERNEL (VCK).** The Strong-Higgs name is retained in the PAPER_2131 append as the historical first canonization (append-only discipline; supersession recorded here, PAPER_2125 Two-Kernel → Two-Layer precedent).

3. The 1/12 tilt factorization — the paper's unification

The session-3 sweep independently found the exact cross-domain identity, published twice for physically distinct observables:

--

$K_{\text{MEX}} - F_{\text{TRZ}} \cdot \Phi_{5/6} = 2$ EXACT (S394: Sommerfeld-Wilson $R_W = 2$, free-electron quantization)

(S418: tokamak edge safety factor $q_{\text{edge}} = 2$)

--

Corollary triple-form (uniting PAPER_1183's DPM-pair identity and PAPER_1156's Hubble tilt):

--

$1/12 = F_{\text{TRZ}} \cdot \Phi_{5/6} = K_{\text{MEX}} - 2$ EXACT

--

The Hubble-tension tilt is the time-reversal-zone fraction times the counting fraction. And substituting PAPER_1522's $K_{\text{MEX}} = \Phi_{5/6} \cdot \text{SO}_{5/D_{\text{phys}}}$ into K :

--

$K = F_{\text{TRZ}} \cdot \Phi_{5/6} \cdot (\text{SO}_{5/D_{\text{phys}}}) \cdot \text{SSq} = (1/12) \cdot (5/2) \cdot (57/100) = 19/160$ EXACT

..

The two sweep findings are one structure: the Vacuum Coupling Kernel factors through the Hubble tilt. The same exact rational $1/12$ that measures the local-vs-CMB expansion discrepancy (PAPER_1156) scales, through the SO_5/D_{phys} dimensional ratio and the source coefficient, into the kernel that carries α_s , λ_H , m_H/m_t , J_{CP} , and N_{eff} . Cosmological tension and particle-sector coupling texture share a common two-primitive core.

4. Honest residuals and disclosures

Instances carry their own published residuals (0.014% – 0.50%; table above); the kernel identity, the rational $K = 19/160$, the tilt decomposition $1/12 = F_{TRZ} \cdot \Phi_{5/6}$, and the factorization $K = (1/12) \cdot (SO_5/D_{phys}) \cdot SSq$ are EXACT arithmetic on published forms (PAPER_1522 + S394/S418). N_{eff} and α_s use the $\Phi_{5/6}$ counting variant per their published dispatch values (bit-verified) — two more counting-sector data points for the PAPER_2129 sector-rule ledger. J_{CP} target 3×10^{10} is the dispatch-stated observation anchor. Discovery method: textual composition matching over published expressions; value-coincidence matching remains rejected (gate-pinned). NOT REPLACEMENT.

5. Falsifiability windows

1. **Sixth VCK instance** among dimensionless couplings/ratios (candidates: $\sin^2\theta_W$ -adjacent forms, y_b/y_t -class ratios, Ω -ratio forms) — by the R400 horizon.
2. **Tilt-factor test:** any future EXACT closure producing $1/12$ should decompose as $F_{TRZ} \cdot \Phi_{5/6}$ (counting sectors) — a wrong-variant appearance ($F_{TRZ} \cdot 0.84 = 0.084 \neq 1/12$) would break the family.
3. **Rational-kernel test:** $K = 19/160$ EXACT means any instance's correction term must carry the full residual vs observation; a future instance requiring a modified kernel value falsifies the census.

6. Cross-references

PAPER_2131 (first canonization + precision route; append superseded by §2); PAPER_1522 ($K_{MEX} = \Phi_{5/6} \cdot SO_5/D_{phys}$); PAPER_1156 ($1/12$ Hubble tilt); PAPER_1183 ($K_{MEX} - 2 = 1/12$ DPM-pair identity); PAPER_1154 (SSq); PAPER_2129 (Φ sector rule); PAPER_2109 (F_{TRZ}^3/F_{TRZ} rungs); sessions S374/S377/S378/S382/S394/S418; dispatch entries listed in §1.

7. Summary Statement

PAPER_2132 canonizes the VACUUM COUPLING KERNEL $K = F_{TRZ} \cdot K_{MEX} \cdot SSq = 19/160$ EXACT at five published instances spanning the strong coupling, the Higgs self-coupling, the Higgs-top mass ratio, the Jarlskog CP invariant, and the cosmological N_{eff} — superseding the provisional Strong-Higgs name — and unifies it with the exact cross-domain identity family $K_{MEX} - F_{TRZ} \cdot \Phi_{5/6} = 2$ (published twice: free-electron R_W and tokamak q_{edge}) whose corollary $1/12 = F_{TRZ} \cdot \Phi_{5/6} = K_{MEX} - 2$ places the Hubble-tension tilt inside the kernel as a literal factor: $K = (1/12) \cdot (SO_5/D_{phys}) \cdot SSq$. The vacuum's coupling texture and the cosmological expansion tension share a common exact two-primitive core.

Filed 2026-07-24. Append-only henceforth.